

YouTube Viral Video Prediction

Using Pre-Publication Information

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Abstract

This project investigates whether a YouTube video’s virality can be predicted using only information available before publication: the video title, publish timestamp, and content category. We define a video as viral if its view count reaches the 90th percentile of the dataset, yielding a binary classification task with roughly 10% positive examples. Two models are compared: a Logistic Regression baseline trained on five structured features, and an RBF Kernel SVM trained on the same structured features augmented with TF-IDF title representations. The SVM achieves an AUC of 0.967 and an F1 score of 0.723, compared to 0.693 and 0.289 for the baseline, demonstrating that title text is the dominant predictive signal. A cross-category generalization experiment further reveals that viral patterns are strongly category-specific, an insight that points toward category-aware modeling as a natural direction for future work.

Keywords: viral prediction, TF-IDF, logistic regression, kernel SVM, PCA

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1 Introduction

Predicting whether a YouTube video will go viral is a problem of practical interest to content creators, platform operators, and marketers. Most approaches to this problem rely on post-publication engagement signals such as view counts, likes, and comment volume — metrics that are only observable after a video has already been released and seen by an audience. This project poses a stricter and more realistic question: *can we predict virality using only information that exists at the moment of upload?*

The features considered are limited to the video title, publish timestamp, and content category, all of which are determined before a video goes live. Engagement metrics (views, likes, dislikes, comment count) are explicitly excluded as model inputs; view counts are used solely for constructing the binary viral label.

Beyond the core prediction task, the project investigates a second question: *do viral patterns generalize across content categories?* A model trained on Gaming videos may have learned patterns specific to that genre. Testing whether those patterns transfer to Music or Entertainment reveals whether virality is a universal phenomenon or a category-specific one — a question with direct implications for how such models should be designed and deployed in practice.

Two main experiments are conducted. Experiment 1 trains a Logistic Regression classifier on structured features only to establish a baseline. Experiment 2 augments these features with a TF-IDF representation of the video title and trains an RBF Kernel SVM, testing whether the specific words in a title carry additional predictive power. Two further experiments visualize the learned feature space via PCA and measure cross-category generalization through a held-out F1 matrix.

2 Data

2.1 Dataset

The dataset used is the **Kaggle Trending YouTube Video Statistics** (US subset), consisting of a CSV file of video records and a JSON category mapping file. It contains metadata for videos that appeared on YouTube’s US trending page, including the video title, publish timestamp, category identifier, view count, like count, dislike count, and comment count. The raw dataset contains 40,949 rows. After removing rows with missing titles or unparseable publish timestamps, the cleaned dataset retains approximately the same volume of samples.

2.2 Label Definition

A binary viral label is constructed from view counts:

$$\text{viral} = \begin{cases} 1 & \text{if views} \geq Q_{0.90}(\text{views}) \\ 0 & \text{otherwise} \end{cases}$$

where $Q_{0.90}$ denotes the 90th percentile of the view count distribution. This yields approximately 10% positive (viral) examples, creating a class imbalance that is addressed during model training.

2.3 Category Distribution

To understand the data, we computed the viral rate for each content category. Categories vary substantially in their fraction of viral videos, as shown in Figure 1, suggesting that category identity carries meaningful signal for prediction.

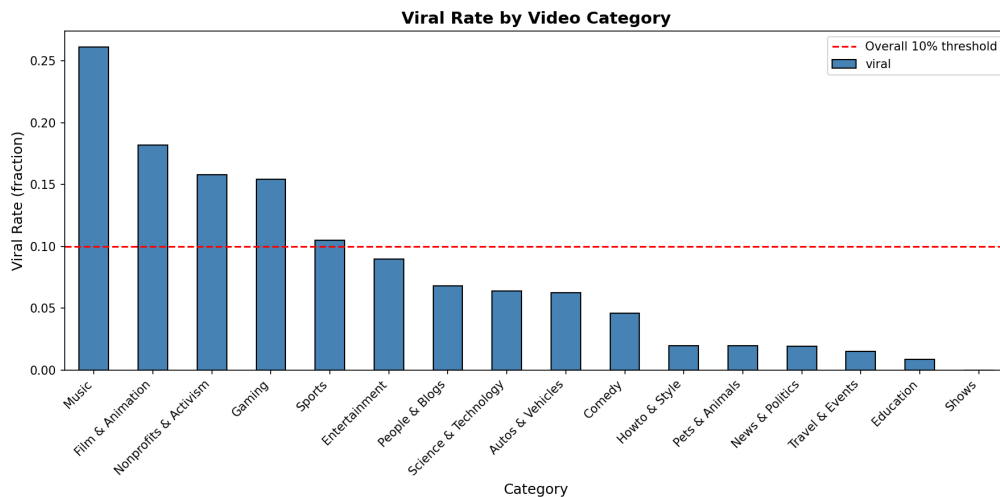


Figure 1: Viral rate broken down by YouTube content category. The dashed red line marks the overall 10% baseline.

3 Methodology

3.1 Feature Engineering

Two feature sets are constructed from the allowed input variables.

Structured features (5 dimensions):

- publish hour — hour of day (0–23) extracted from the publish timestamp
- publish weekday — day of week (0 = Monday, 6 = Sunday)
- title length — number of characters in the title
- title word count — number of whitespace-delimited tokens in the title
- category id — integer category identifier

Text features: A TF-IDF representation of the video title is computed with a vocabulary limited to the top 500 terms (English stop words removed), yielding a 500-dimensional sparse vector per video.

The full feature matrix concatenates the scaled structured features and the TF-IDF vector, producing a 505-dimensional representation. To prevent data leakage, the standard scaler for structured features, the TF-IDF vectorizer, and the PCA transform used in Experiment 2 are all fitted exclusively on the training split and then applied to the test split.

3.2 Train/Test Split

The dataset is divided into a 70% training set and a 30% test set using stratified sampling (preserving the viral/non-viral ratio in both splits) with a fixed random seed for reproducibility.

3.3 Experiment 1 — Logistic Regression Baseline

Logistic Regression with ℓ_2 regularization is trained on the structured features only. The model minimizes the regularized cross-entropy loss:

$$\min_{w,b} \sum_{i=1}^n \log(1 + e^{-y_i(w^\top x_i + b)}) + \frac{1}{2C} \|w\|^2$$

where $C > 0$ controls regularization strength (larger C = less regularization). Class imbalance is handled by reweighting each training example inversely proportional to its class frequency. The optimal C is selected via 5-fold stratified cross-validation over the grid $C \in \{0.001, 0.01, 0.1, 1, 10\}$, with F1 score as the selection criterion.

3.4 Experiment 2 — RBF Kernel SVM

A Support Vector Machine with the Radial Basis Function (RBF) kernel is trained on the full feature set. Because the SVM's computational cost scales poorly with the number of features, PCA first reduces the 505-dimensional feature matrix to 100 principal components. The RBF kernel implicitly maps inputs into a high-dimensional feature space:

$$K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2)$$

The SVM then finds the maximum-margin separating hyperplane in this space. Hyperparameters $C \in \{0.1, 1, 10\}$ and $\gamma \in \{\text{scale}, 0.01, 0.1\}$ are jointly tuned via 5-fold cross-validation optimizing F1. Class imbalance is handled identically to Experiment 1.

3.5 Experiment 3 — PCA Visualization

To examine the geometry of the feature space, PCA is applied to project the full training feature matrix to 2 dimensions. Viral and non-viral samples are plotted separately to assess visual separability, both for the full training set and individually for four categories: Gaming, Music, Entertainment, and Science & Technology.

3.6 Experiment 4 — Cross-Category Generalization

To assess whether viral patterns transfer across content genres, the four most frequent categories are selected. For each ordered pair (train category, test category), a separate SVM is trained on all samples from the training category and evaluated on all samples from the test category using the best hyperparameters from Experiment 2. This produces a 4×4 F1 matrix whose diagonal entries represent in-domain performance and whose off-diagonal entries represent cross-domain generalization.

4 Implementation Details

All experiments are implemented in Python 3 using scikit-learn. Table 1 summarizes the key hyperparameter settings.

Table 1: Implementation settings used across all experiments.

Setting	Value
Train / test split ratio	70 / 30, stratified
Random state	42 (all steps)
TF-IDF vocabulary size	500
TF-IDF stop words	English
PCA components (Exp. 2)	100
LR solver	L-BFGS, max iterations = 1000
LR C grid	{0.001, 0.01, 0.1, 1, 10}
SVM kernel	RBF
SVM C grid	{0.1, 1, 10}
SVM γ grid	{scale, 0.01, 0.1}
Cross-validation folds	5-fold stratified
CV scoring metric	F1
Class imbalance handling	Inverse-frequency sample weighting
Leakage prevention	Scaler / TF-IDF / PCA fit on training set only

5 Results

5.1 Experiment 1 — Logistic Regression Baseline

We trained a Logistic Regression model using only the five structured features and evaluated it on the held-out test set. The best regularization parameter found by cross-validation was $C = 0.1$. Inspecting the learned coefficient vector revealed that category id and publish hour carry the largest absolute weights, suggesting that category membership and publishing time encode the most useful signal among the structured features.

The model achieves a recall of 0.550, meaning it identifies roughly half of all viral videos, but precision is low at 0.196, indicating a high false-positive rate. The AUC of 0.693 is above random chance but leaves considerable room for improvement. These results establish the

ceiling of what structured features alone can offer, motivating the addition of text features in Experiment 2. Figure 2 shows the ROC curve and confusion matrix.

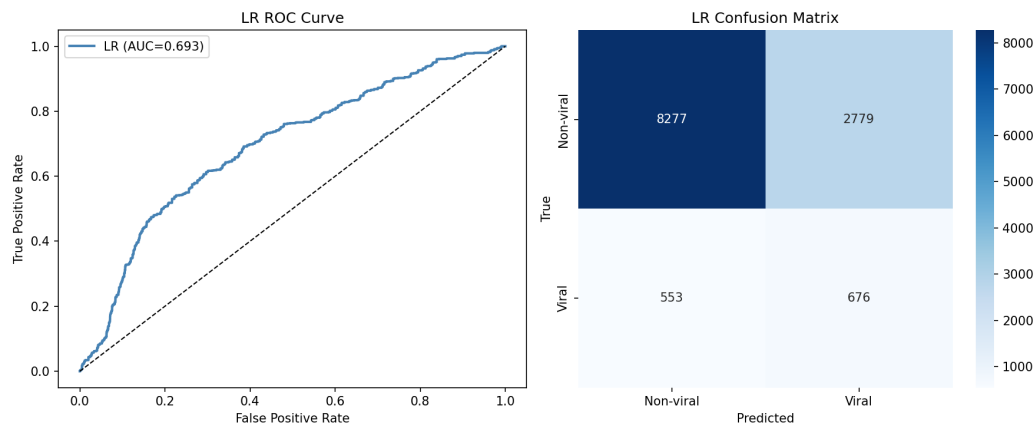


Figure 2: Logistic Regression baseline results: ROC curve (left) and confusion matrix (right) on the held-out test set.

5.2 Experiment 2 — Kernel SVM

We augmented the structured features with TF-IDF title representations, applied PCA to reduce dimensionality to 100 components, and trained an RBF Kernel SVM. Cross-validation selected $C = 10$ and $\gamma = \text{scale}$ as the best hyperparameters.

The results, summarized in Table 2, show strong improvement across every metric. F1 increases by 150.5% relative to the baseline, and AUC reaches 0.967. Recall of 0.937 means the model correctly identifies 93.7% of all truly viral videos. Precision improves from 0.196 to 0.589, indicating that most videos predicted as viral do indeed go viral. These results demonstrate that the words in a video’s title are a powerful predictor of virality — far more informative than timing and category alone.

Table 2: Test set performance of the two models.

Model	Accuracy	Precision	Recall	F1	AUC
LR Baseline (structured only)	0.729	0.196	0.550	0.289	0.693
Kernel SVM (structured + TF-IDF)	0.928	0.589	0.937	0.723	0.967
Absolute improvement (Δ)	+0.199	+0.393	+0.387	+0.434	+0.274
Relative improvement (%)	+27.3%	+200.8%	+70.4%	+150.5%	+39.6%

Figure 3 provides the ROC comparison, SVM confusion matrix, and a side-by-side metric bar chart for both models.

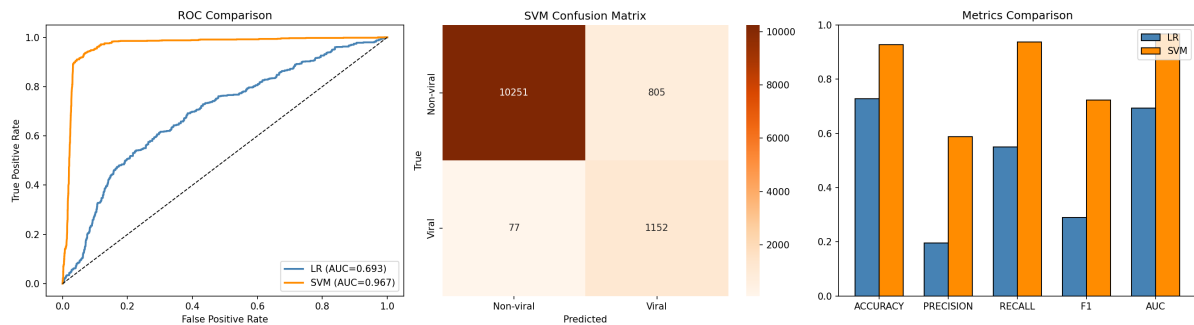


Figure 3: Experiment 2 results: ROC curve comparison (left), SVM confusion matrix (center), and per-metric bar chart comparing LR and SVM (right).

5.3 Experiment 3 — PCA Visualization

To gain intuition about the feature space, we projected the 505-dimensional training features down to 2 principal components and plotted viral versus non-viral samples. While the two classes overlap to some degree — expected given the inherent difficulty of the task — viral videos occupy a noticeably distinct region along the first principal component, confirming that the learned features carry genuine discriminative signal.

The per-category breakdown, shown in Figure 5, reveals meaningful variation across genres. Table 3 reports the Euclidean distance between the viral and non-viral cluster centroids for each category. Science & Technology and Gaming exhibit the clearest separation, while Music videos cluster together regardless of virality, suggesting that title-based features are more informative for some genres than others.

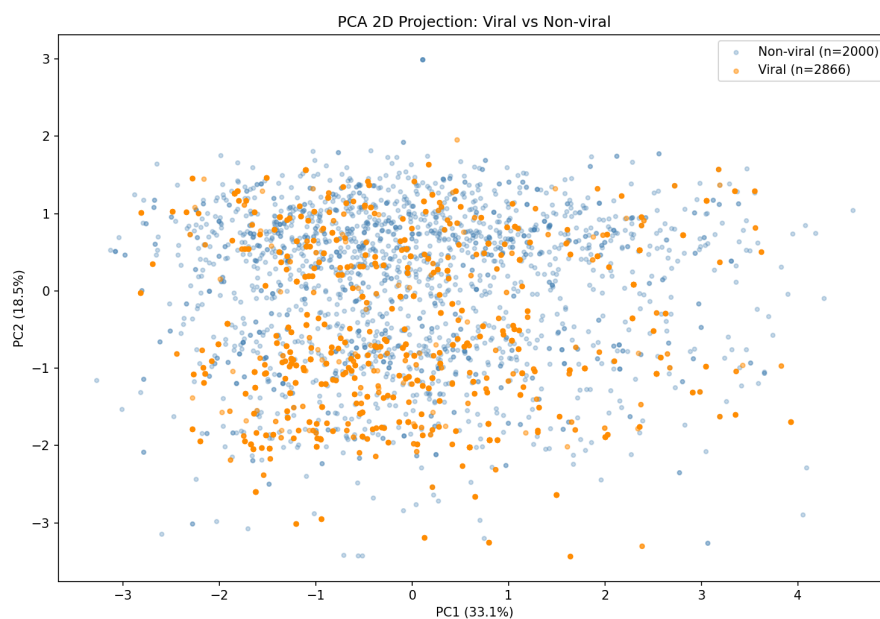


Figure 4: PCA 2D projection of the full training set. Orange points are viral videos; blue points are a random subsample of 2,000 non-viral videos.

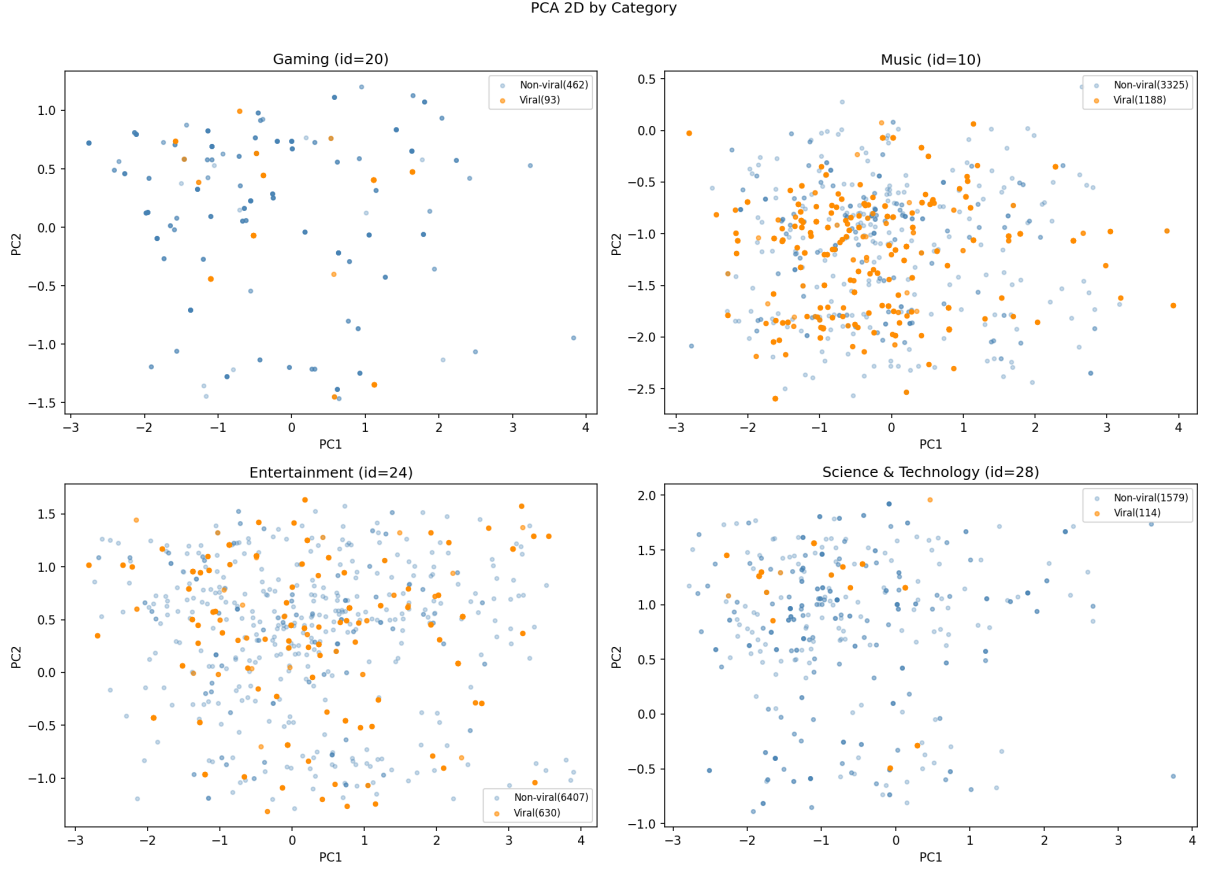


Figure 5: Per-category PCA 2D projections for Gaming, Music, Entertainment, and Science & Technology.

Table 3: Centroid distance between viral and non-viral clusters in 2D PCA space, by category.

Category	Centroid Distance	Separation
Science & Technology	0.672	Clear
Gaming	0.454	Clear
Entertainment	0.202	Moderate
Music	0.112	Weak

5.4 Experiment 4 — Cross-Category Generalization

To understand how well viral patterns transfer across genres, we trained separate SVM models on each of the four most frequent categories and tested each model on every category, yielding the 4×4 F1 matrix in Figure 6. The diagonal entries (in-domain) average 0.555, confirming that category-specific models learn meaningful patterns within their own genre. The off-diagonal entries (cross-domain) average only 0.049, a gap of 0.506.

This large gap reveals an important structural property of the data: viral patterns are largely category-specific. The language and framing that makes a Gaming title go viral differs substantially from what works in Music or Entertainment. Music is the most category-specific

genre in this respect — models trained on Music generalize least to other categories, consistent with the weak PCA separation observed in Experiment 3.

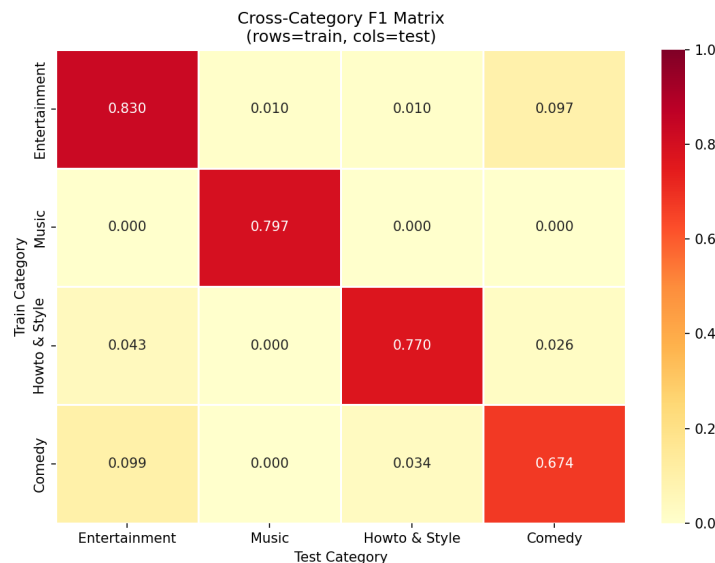


Figure 6: Cross-category F1 matrix. Rows indicate the training category; columns indicate the test category. Diagonal entries are in-domain results; off-diagonal entries are cross-domain results.

6 Discussion

Viral video prediction from pre-publication features is achievable. The core finding of this project is that virality can be predicted with high accuracy before a video is published. The Kernel SVM achieves an AUC of 0.967 and an F1 of 0.723 — strong performance on a task that uses no engagement data whatsoever. This validates the premise that meaningful signals about a video’s potential are encoded in its title, publish time, and category.

Title text is the dominant signal. The 150.5% relative improvement in F1 from the structured-only baseline to the text-augmented SVM shows that the specific words a creator chooses carry far more predictive power than timing or category alone. This is an actionable finding: content creators can use title optimization as a concrete lever for improving reach.

Viral patterns are category-specific, which is itself a useful finding. The cross-category experiment was designed to ask whether a single global model would be sufficient. The answer is no — but this is not a failure of the model. It is a discovery about the nature of virality on YouTube: different genres have different linguistic cultures, and what goes viral in one does not necessarily go viral in another. Knowing this guides how future systems should be built — with separate, category-aware models rather than a single one-size-fits-all classifier.

Practical implications. These findings translate into concrete guidance for content creators. First, since title text is the strongest predictive signal, investing effort in title wording — choosing specific, evocative language rather than generic descriptions — is likely to have the greatest

impact on a video’s reach. Second, because publish hour and weekday are among the most informative structured features, timing uploads strategically (e.g., aligning with peak audience activity periods) provides an additional marginal advantage. Third, the category-specific nature of viral patterns means that creators should study what works within their own genre rather than emulating strategies from unrelated categories — a title style that performs well in Science & Technology may be counterproductive in Music. Finally, the trained model itself can serve as a practical pre-upload screening tool: given a candidate title, publish time, and category, it can estimate the probability of going viral before a video is ever released, enabling iterative title refinement at zero cost.

Future directions. Building on these results, natural extensions include training dedicated per-category models, exploring richer text representations such as pretrained sentence embeddings, and incorporating additional pre-publication signals such as channel subscriber count or thumbnail features.

7 Conclusion

This project shows that predicting YouTube video virality from pre-publication information alone is not only feasible but achievable at a high level of performance. By combining structured metadata with TF-IDF title features and training an RBF Kernel SVM, we reach an AUC of 0.967 and an F1 score of 0.723 on a held-out test set — a 150% improvement in F1 over the structured-only Logistic Regression baseline. The title text of a video is the strongest available pre-publication signal, contributing the majority of the predictive gain.

The cross-category generalization experiment adds a deeper insight: viral patterns are not universal but genre-dependent, with in-domain F1 averaging 0.555 and cross-domain F1 averaging 0.049. Rather than undermining the project’s success, this finding clarifies the right modeling strategy going forward — category-aware models that respect the distinct linguistic norms of each genre. Together, these results demonstrate that machine learning can extract meaningful structure from simple pre-publication metadata, with practical implications for content strategy and platform design.